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The Editors and Guest Editors
Special Collection: Statistical Foundations of AI and Real-World Evidence Generation
Biostatistics

Dear Editors,

We submit our manuscript "Bayesian fusion forests for heterogeneous treatment effects on survival from randomised and real-world data" for consideration in the special collection on Statistical Foundations of AI and Real-World Evidence Generation.

Randomised trials identify treatment effects but are underpowered for subgroup effects and often stop before the events of interest accumulate. Real-world data are larger and follow patients for longer, but treatment there is not randomised. Our manuscript develops the Bayesian fusion forest, a nonparametric framework that combines the two sources without assuming the observational source is unconfounded. A confounding function absorbs the bias from unmeasured confounding, a shared baseline prognosis with a source-specific deviation encodes the borrowing, and a hierarchical Dirichlet process mixture lets the error distribution differ between the sources. Each component carries an independent causal meaning and receives its own tree-ensemble prior. To our knowledge the framework is the first that combines causal effect estimation, data fusion and support for both right- and interval-censored outcomes.

The work speaks to several themes of the collection. It offers a statistical perspective on combining experimental and real-world data, and it performs causal inference with a machine-learning model for the purpose of real-world evidence generation. Transportability across data sources is the assumption that makes the fusion possible, and we state its cost explicitly. Uncertainty quantification is intrinsic rather than added afterwards, and our cohort records outcomes only at annual visits, so the event times are interval-censored, one of the data imperfections the call names.

A simulation study shows efficiency gains over a trial-only analysis across varying levels of confounding and between-source heterogeneity. We benchmark against four machine-learning alternatives, and the Bayesian fusion forest attains a lower error than every alternative in every configuration while remaining the only method whose intervals reach nominal coverage at a competitive width. We apply the method to the ACTG 175 trial combined with the Multicenter AIDS Cohort Study, where the fusion identifies a benefit of combination antiretroviral therapy for nearly every patient while the trial alone is inconclusive.

The manuscript is original, is not under consideration elsewhere, and all authors have approved the submission. The authors declare no conflict of interest. The software implementation, together with the code for the simulation studies and the data analysis, is available at https://github.com/tijn-jacobs/FusionForests.

The authors used generative artificial intelligence tools in support of developing and debugging the accompanying code. All such output was reviewed and verified by the authors, who take full responsibility for the content of this work.

Yours sincerely,

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